

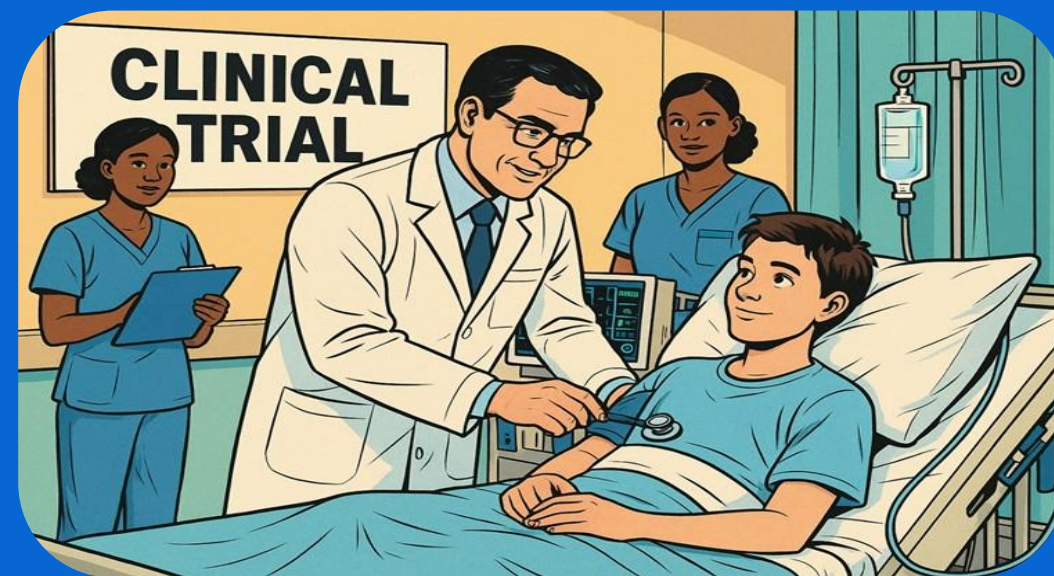


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Life Science Use Case

A Practical Hands-On Workshop using
SAS for a Clinical Trial Analysis



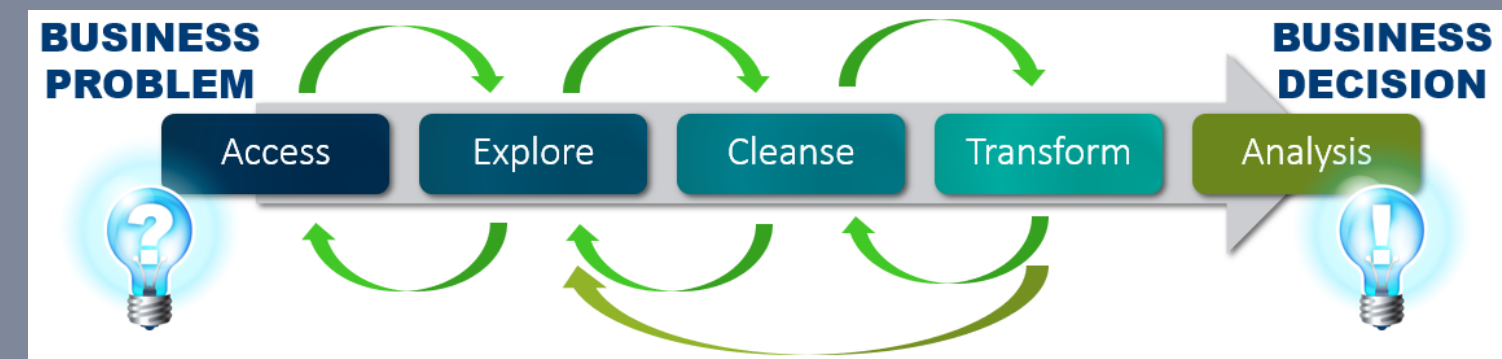
Clinical Trial Analysis



This use case is designed for beginners to learn the basic and essential steps for analyzing Clinical Trials with the enterprise SAS software in a highly regulated environment. No prior knowledge, e.g. SAS, Programming, is necessary. All analysis are created using a no-code, low-code approach.

The Use Case includes different parts along the typically steps of a Data Science Project:

- Access / Loading Data
- Investigate / Clean / Transform Data
- Reporting / Dashboards
- Descriptive Analysis
- Creating Analytical Models
- Survival Analysis



With the skills learned in this hands-on workshop, participants will then be able to perform analysis independently in SAS Visual Analytics. Students use the free of charge SAS Skill Builder e-learning and the included SAS Viya for Learners environment. Working through the use case takes approximately 90 - 120 minutes.

Clinical Trial Analysis



Business Context

You are a biostatistician and you work for an important life science company, which is about to launch a new product SASULIN on the market.

On that regard, the company need to conduct a clinical trial to ensure our products safety and approval by the regulating authorities.

In your role as a biostatistician, with no programming skills, your task is to analyze the given data from the clinical trial and answer important questions about the safety of the new product SASULIN.

Data Dictionary



Four data sets, contains several information and observations about the patients who took part in the clinical trial. The data are available in ADAM and CDISC SDTM format, a data format established in the clinical environment.

Table: ADTTE

Description:

26 Variables, 254 Rows

Excerpt

Variable	Description
AVAL	is the elapsed time to the event of interest from the origin. For example, if AVAL is measured in days, AVAL would be $ADT - STARTDT$ or $ADT - STARTDT + 1$
CNSR	is a required variable for a time-to-event analysis dataset, though it is a conditionally required variable with respect to the ADaM BDS. For example, $CNSR = 0$ for event and $CNSR > 0$ for censored records.
TRTP	is a record-level identifier that represents the planned treatment attributed to a record for analysis purposes.

Data Dictionary



Table: DM_DM

Description: Demographic Data

28 Variables, 60 Rows

Excerpt

Variable	Description
USUBJID	Patient ID
ARM / ARMCD	planed treatment the subject taken (ARMCD if you want to use the code instead of the label)
ACTARM/ACTARMCD	actual treatment the subject taken (ACTARMCD if you want to use the code instead of the label)
RACE	Race
SEX	Gender
SITEID	Site Number

Data Dictionary



Table: AE_DM

Description: Adverse Events

54 Variables, 40 Rows

Excerpt

Variable	Description
RFSTDTC / RFENDTC	indicate a patient’s initial and final exposure to a study drug or treatment, crucial for analyzing patient involvement and events within the study. Understanding the patient’s study involvement timeline is crucial for event analysis.
RFXSTDTC / RFXENDTC	variables specifically represent the date and time of the first and last study exposure to the protocol-specified treatment or therapy, respectively.
RFICDTC	The earliest date of informed consent among any subject (Date/Time of Informed Consent) that enrolled in the study.
RFPENDTC	Final date on which data was collected for any subject (Date/Time of End of Participation, RFPENDTC) based on the protocol.
PATIENTID	Patient ID
AGE	Age
SEX	Gender
AESEV	Severity
AEBODSYS	The body system (also know as system organ class) that is affected.
AEHLGT	coded using the Medical Dictionary for Regulatory Affairs (MedDRA). Sponsors should code each term to the MedDRA-specified hierarchy of terms
AEDECOD	dictionary-derived text description; equivalent to the Preferred Term (PT in MedDRA). The sponsor is expected to provide the dictionary name and version used to map the terms utilizing the define.xml external codelist attributes.
AEREL	Adverse Event Relativity

Data Dictionary



Table: DIABETES

Description: Linked patient data between treatment and Diabetes
9 Variables, 768 Rows

Variable	Description
OUTCOME	Binary, does the drug have an impact on the patient.
AGE	Age
BloodPressure	Blood Pressure
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes Pedigree
Glucose	Glucose
Insulin	Insulin
Pregnancies	Pregancy
SkinThickness	Skin thickness

Clinical Trial Analysis



Goal / Questions

Reporting of the Patient Demography Distribution

- How many patients are engaged / enrolled in that clinical trial ?
- What is the number of patients by type of drugs?
- What is the split of patient by age category?
- How is gender impacting the number of patients?
- What are the impacts of race or site number?

Adverse Events this Clinical Trial is potentially causing

- Which patients facing an Adverse Event?
- How does the event are across?
- What is the investigator's opinion as to the causality of the event to the treatment?
- What Adverse Event's are qualified as serious or not?

Prediction of impact of patients

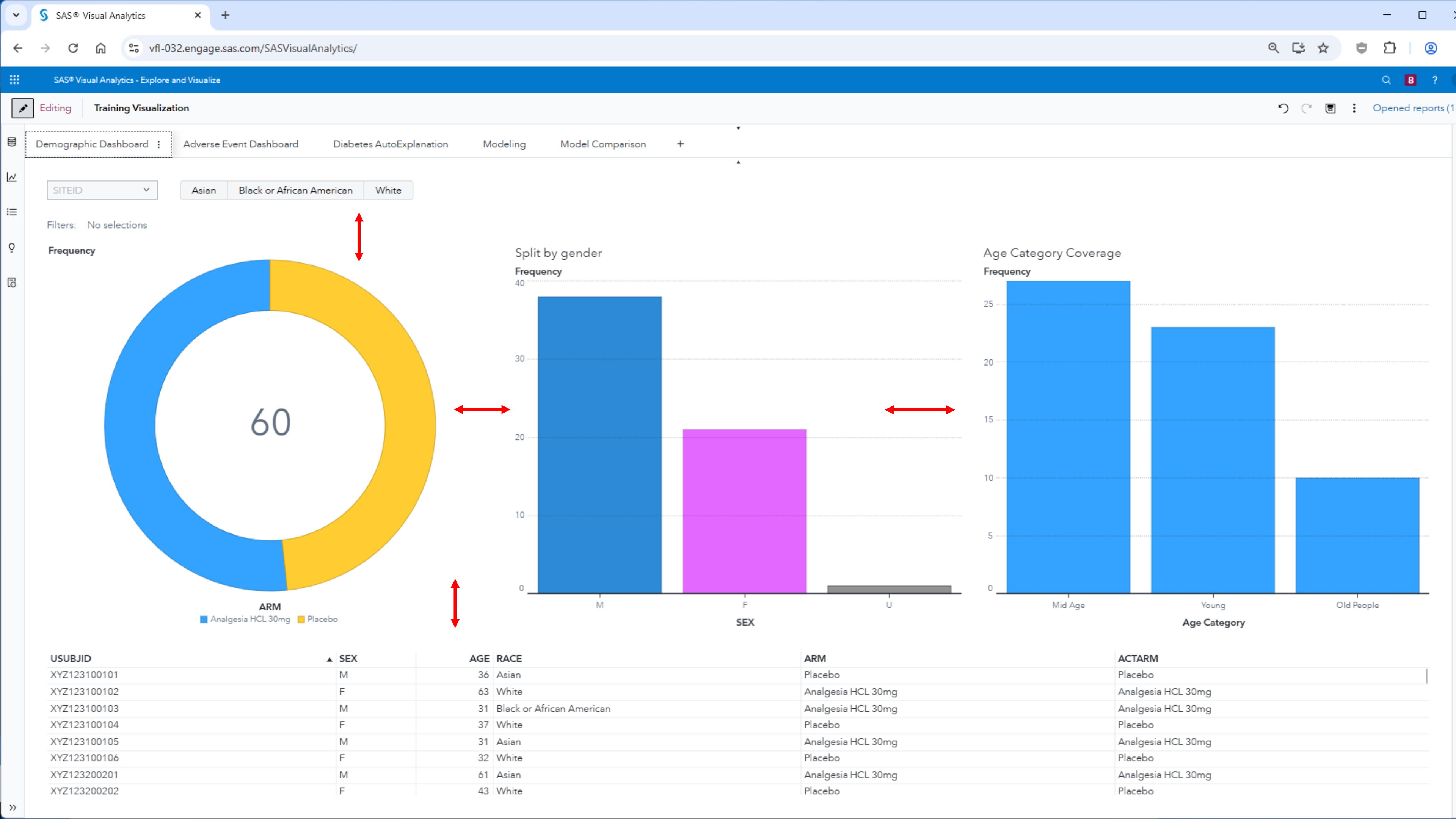
- What are the main factors of impact?

Survial Analysis

- What is the proportion of a population which will survive past a certain time?
- Of those that survive the main event, at what rate will they die?
- Can multiple causes of death be taken into account?
- How do particular circumstances or characteristics increase or decrease the probability of survival?

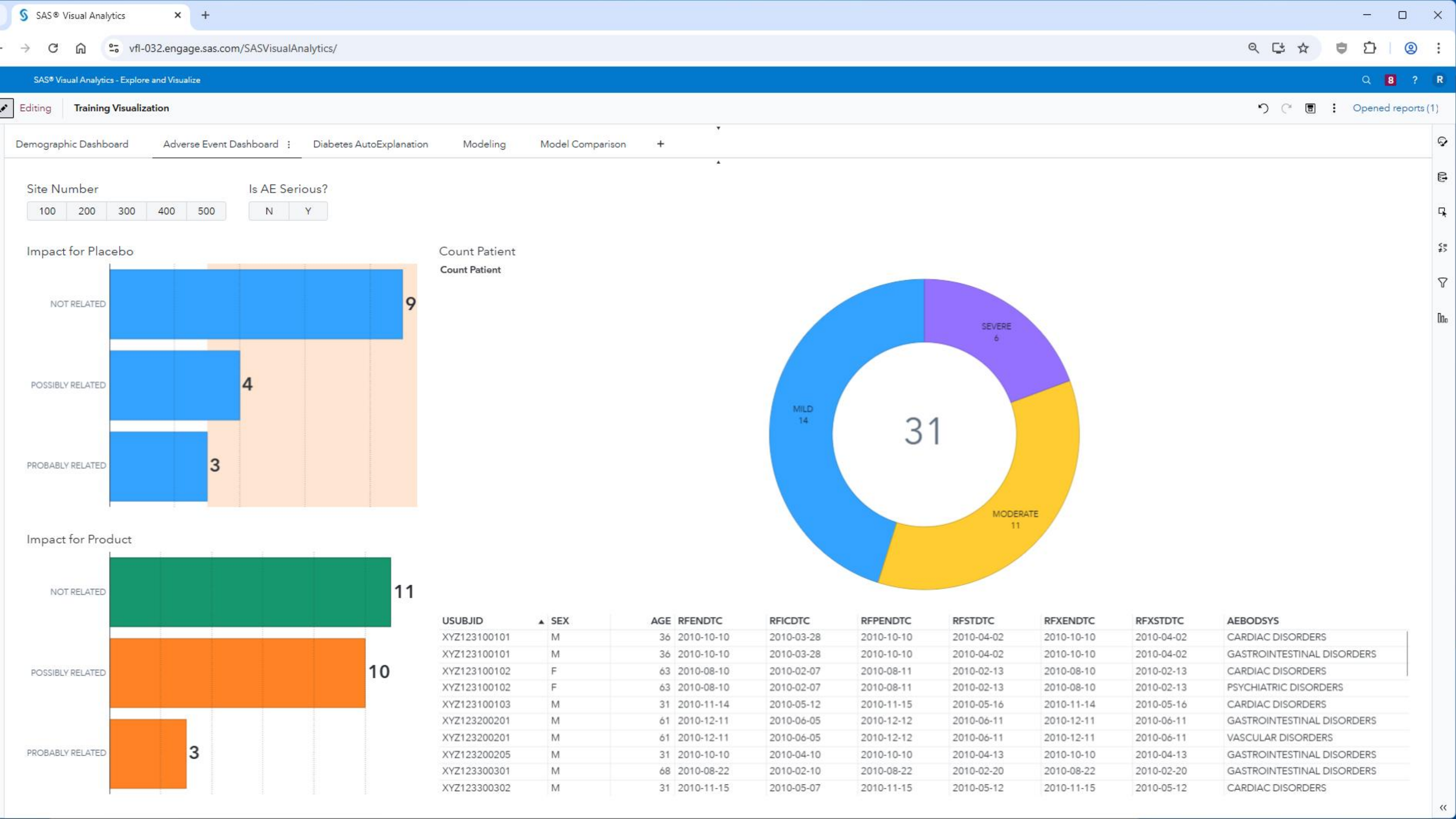
Demographic Dashboard





Adverse Event Dashboard

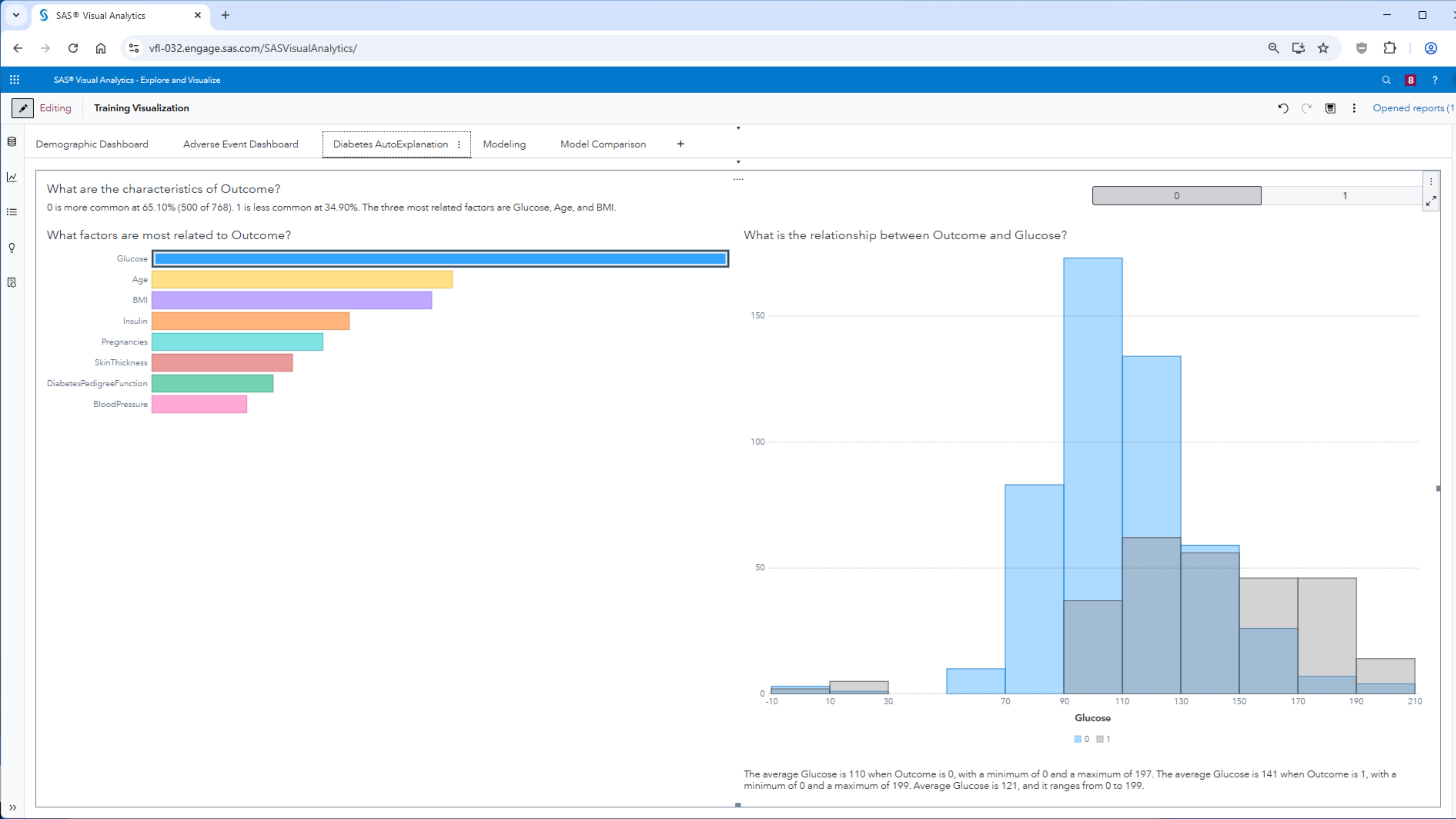




Prediction for patients with Diabetes

Automated-Explanation, Logistic Regression, Forest, Decision Tree

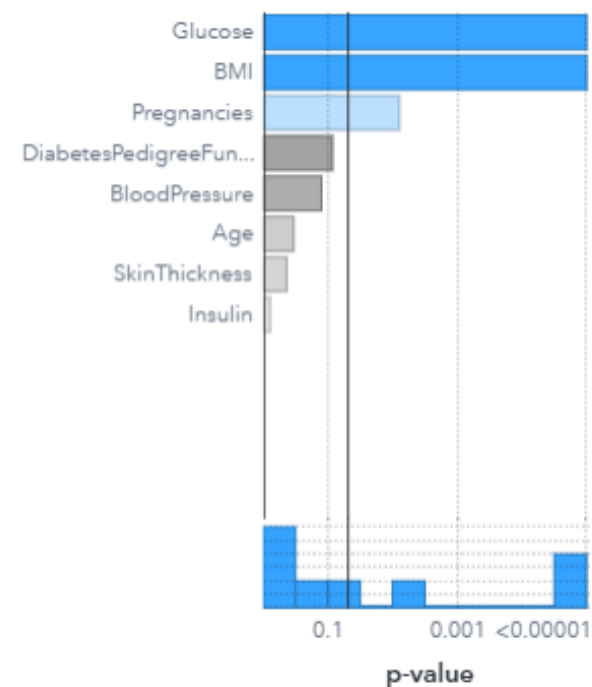




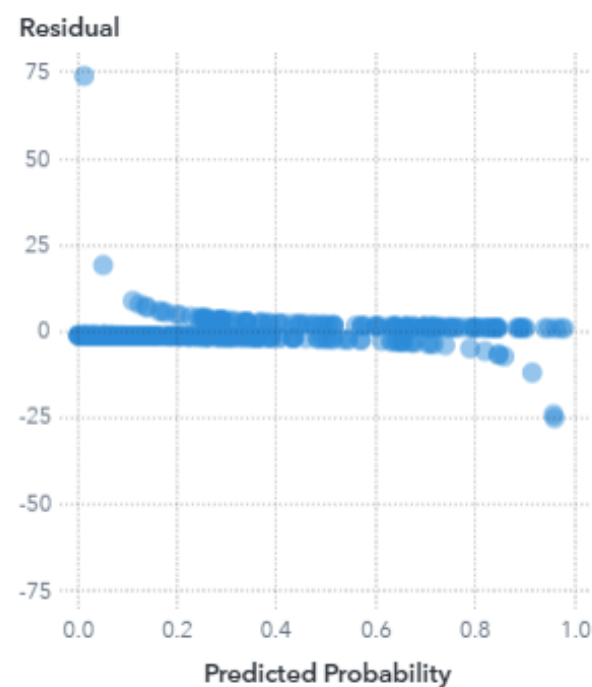
Logistic Regression of Outcome

Event: 1 ▾ Fit: Validation KS (Youden) 0.4894 ▾ Observations: 768 of 768

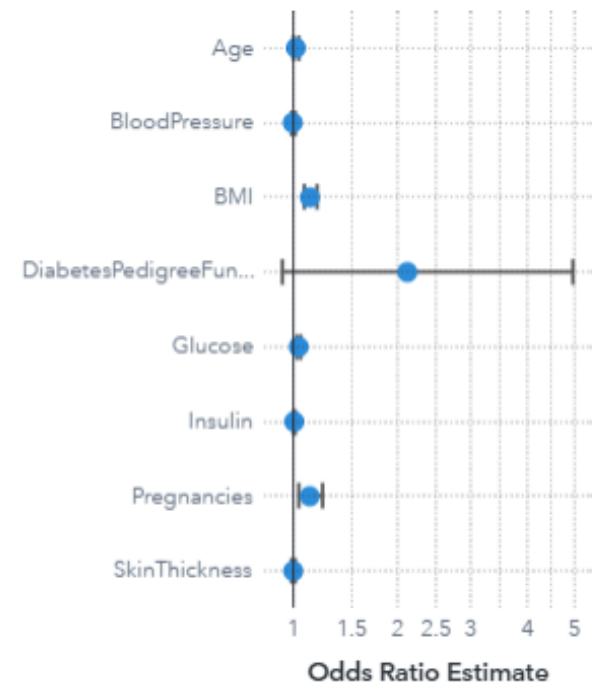
Fit Summary



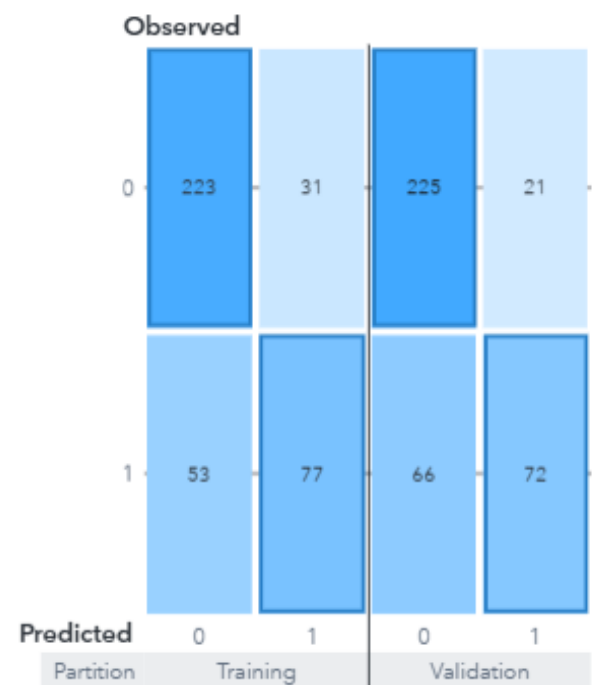
Residual Plot



Odds Ratio Plot ①



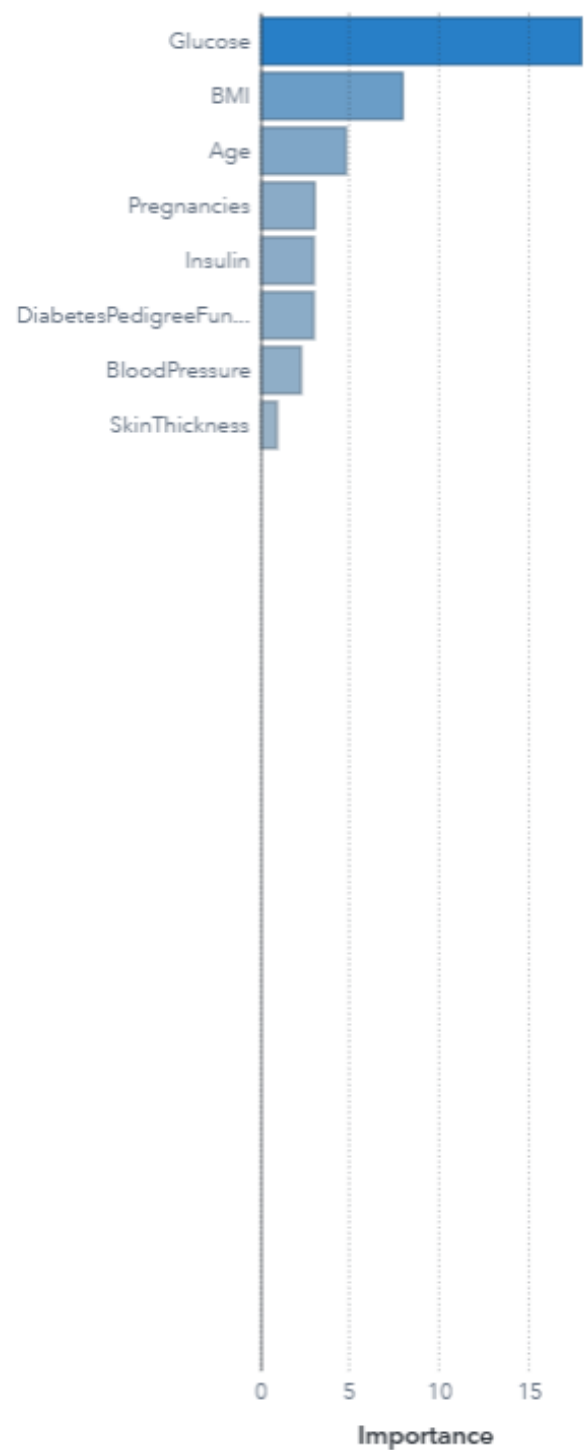
Confusion Matrix ①



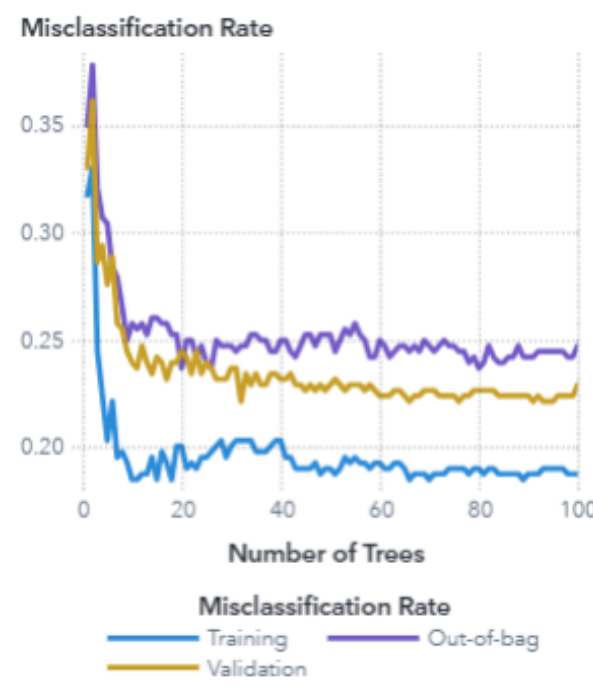
Forest of Outcome

Event: 1 ▾ Fit: KS (Youden) 0.6420 ▾ Observations: 768 of 768

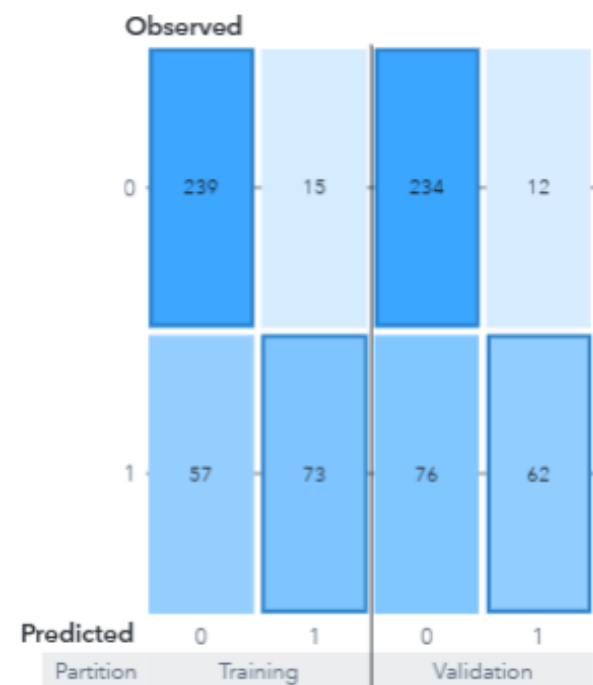
Variable Importance



Error Plot



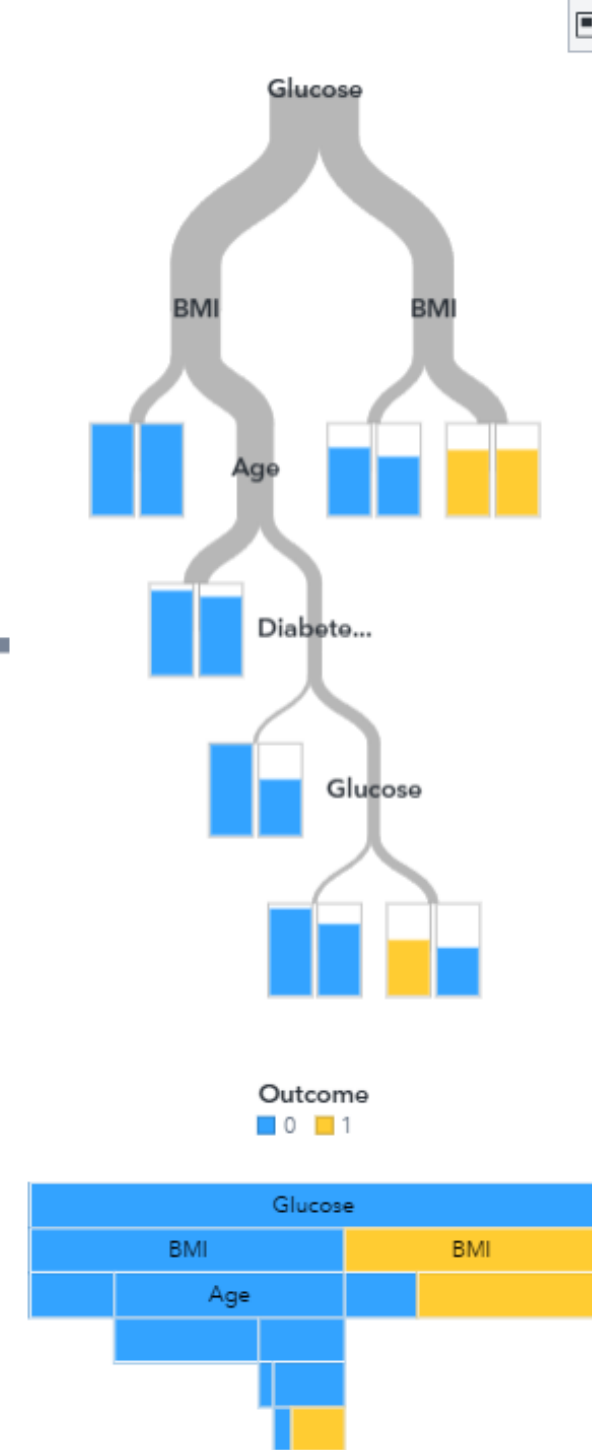
Confusion Matrix



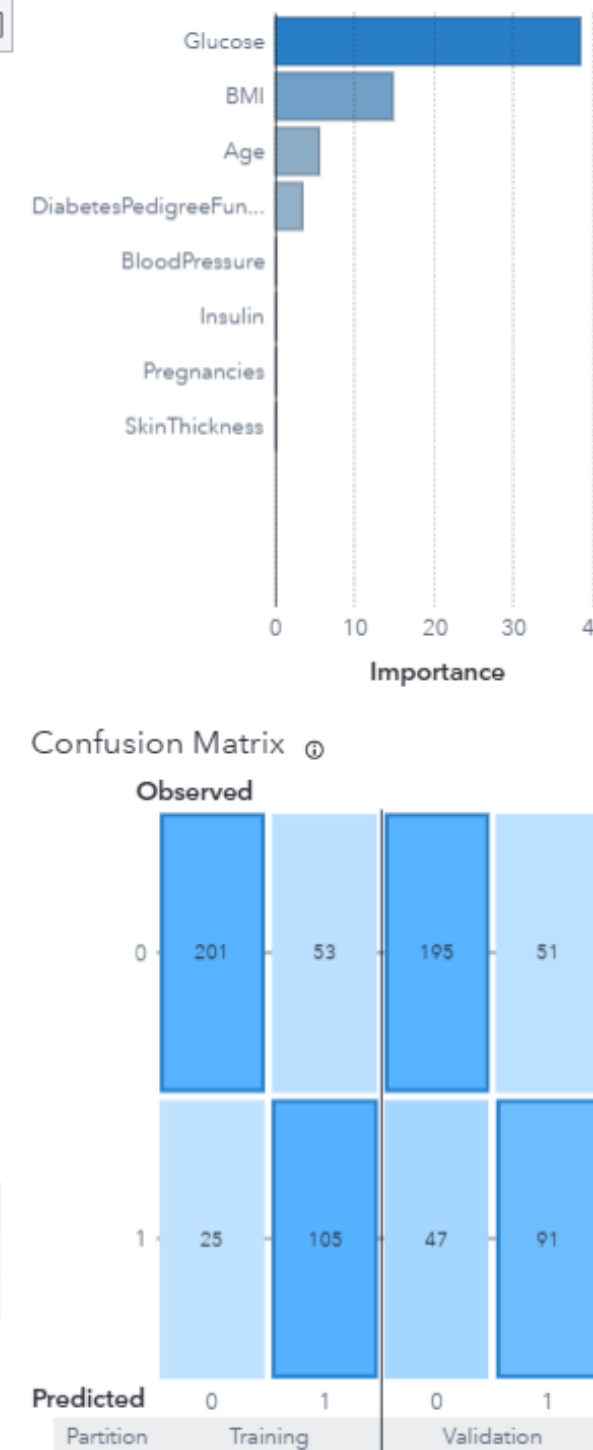
Decision Tree of Outcome

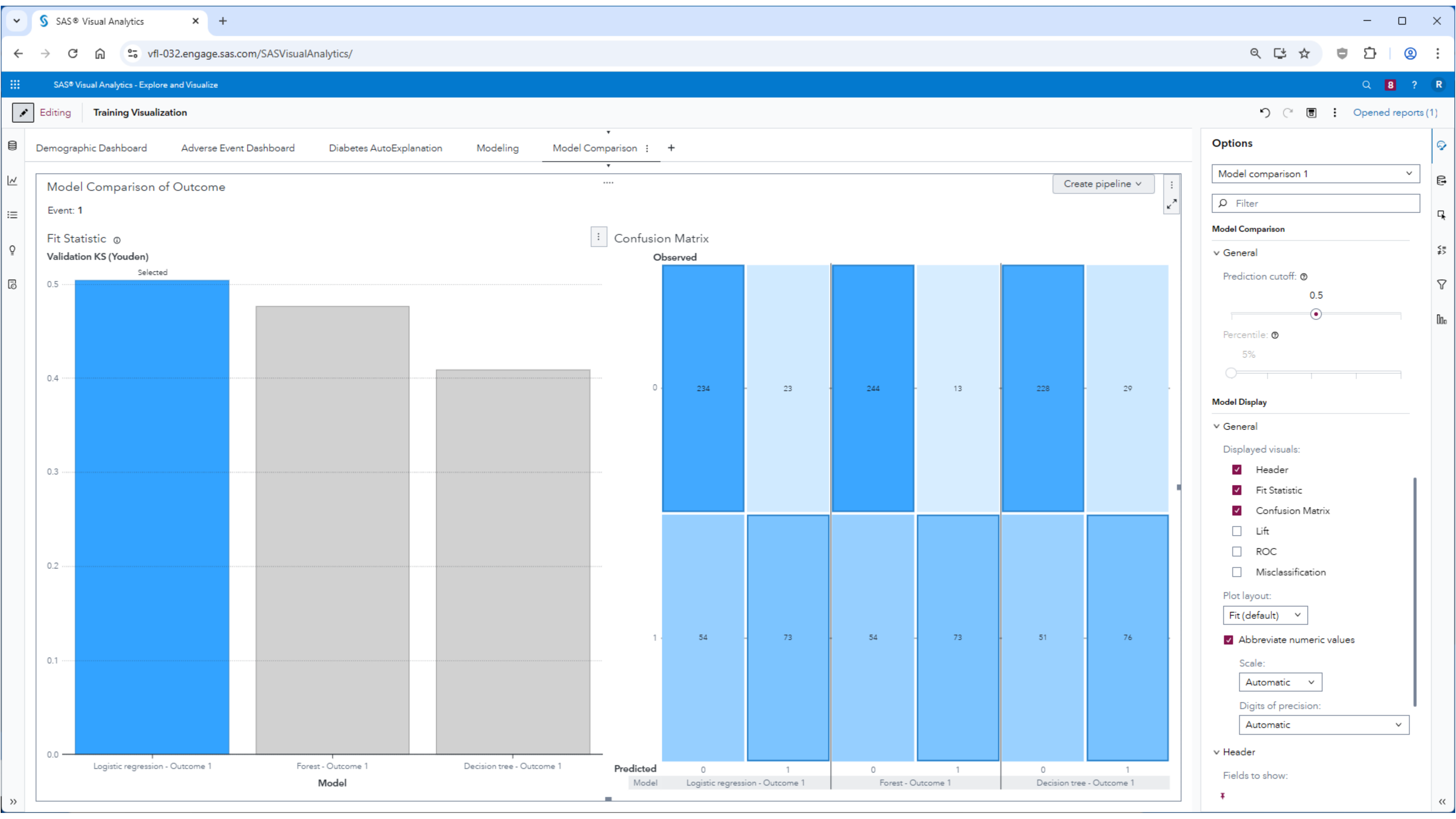
Event: 1 ▾ Fit: Validation KS (Youden) 0.4647 ▾ Observations: 768 of 768

Tree



Variable Importance





Advanced Modeling

Next future steps, first view - Pipeline Building





< Projects



Diabetes Prediction



Data

Pipelines

Pipeline Comparison

Insights



Pipeline 1



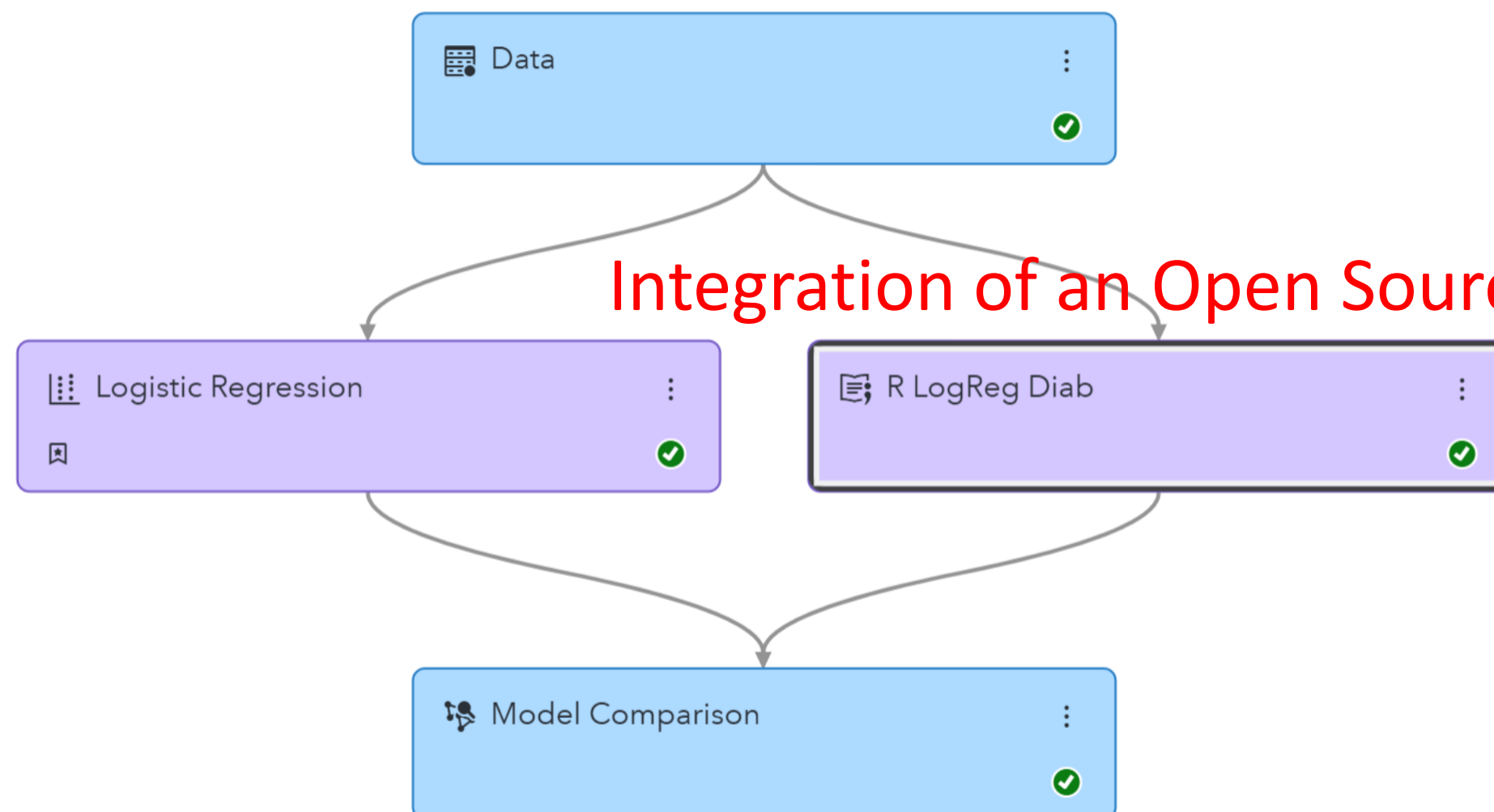
Basic Model



1



Run pipeline



Integration of an Open Source R model

R LogReg Diab



Description:

Runs Python or R code.

Open code editor

Language:

R



v Input to Open Source

> Data Sample

☒ Generate data frame☒ Use output data in child nodes☒ Use the exact percentile method for lift calculations



< Projects



Diabetes Prediction



Data

Pipelines

Pipeline Comparison

Insights



Basic

Intermediate

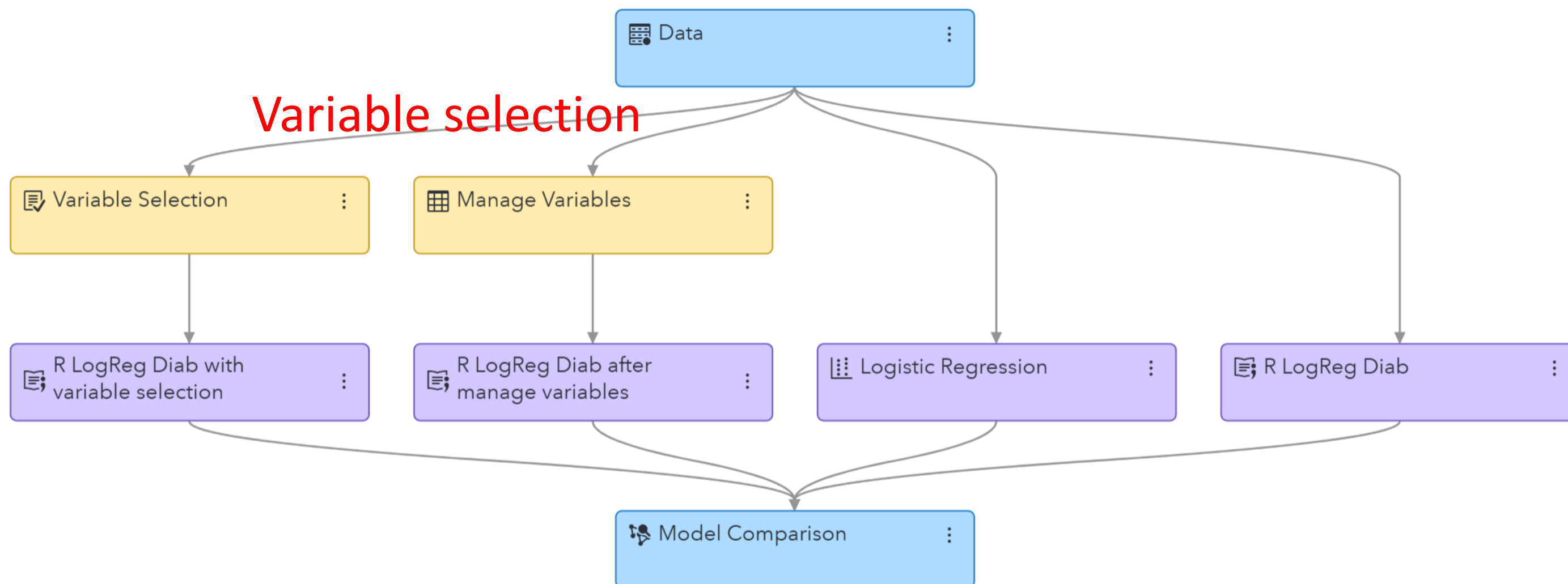
Advanced



Run pipeline



Variable selection





< Projects



Diabetes Prediction



Data

Pipelines

Pipeline Comparison

Insights



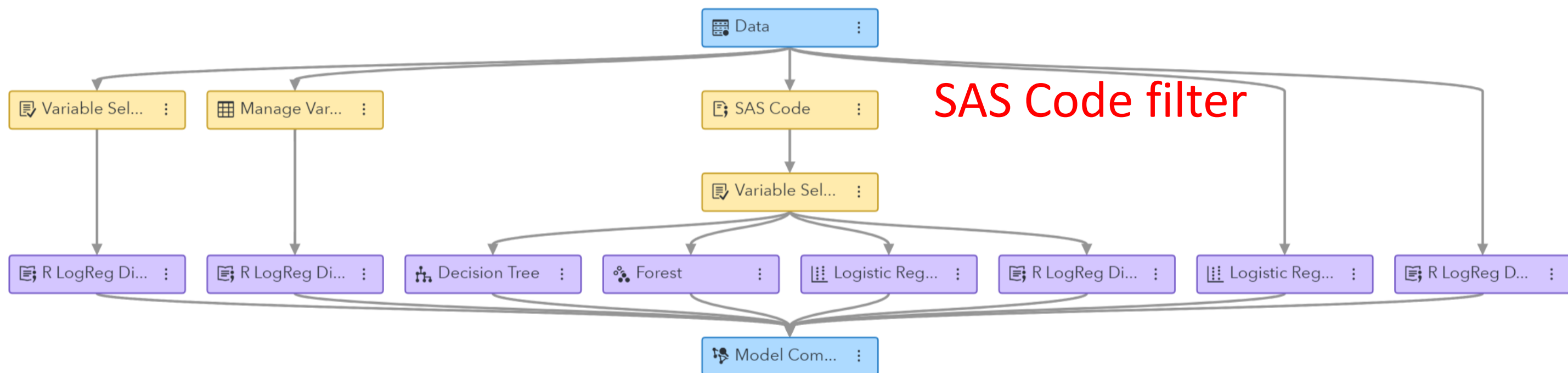
Basic

Intermediate

Advanced



Run pipeline



Model Studio - Build Models

Diabetes_prediction > "R_logreg_diab (1)" Results

"R_logreg_diab (1)" Results

SummaryOutput Data

NodeAssessment

R Code

```
1 #-----
2 # Language: R
3 #-----
4 # Set R-work to node-work directory
5 dm_nodedir <- "/opt/sas/viya/config/var/tmp/compsrv/default/ee5d28e6-9858-4b78-90da-ef6f9f8be09f/SAS_u
6 setwd(dm_nodedir)
7
8 # Variable declarations
9 dm_dec_target <- 'Outcome'
10 dm_partitionvar <- '_PartInd_'
11 dm_partition_train_val <- 1
12
13 dm_class_input <- c()
14 dm_interval_input <- c("DiabetesPedigreeFunction","Glucose","Insulin","Pregnancies","SkinThickness")
15 dm_input <- c(dm_class_input, dm_interval_input)
16 dm_model_formula <- as.formula(paste(dm_dec_target, '~', paste(dm_input, collapse='*')))
17 dm_predictionvar <- c('P_Outcome0', 'P_Outcome1')
```

R Output

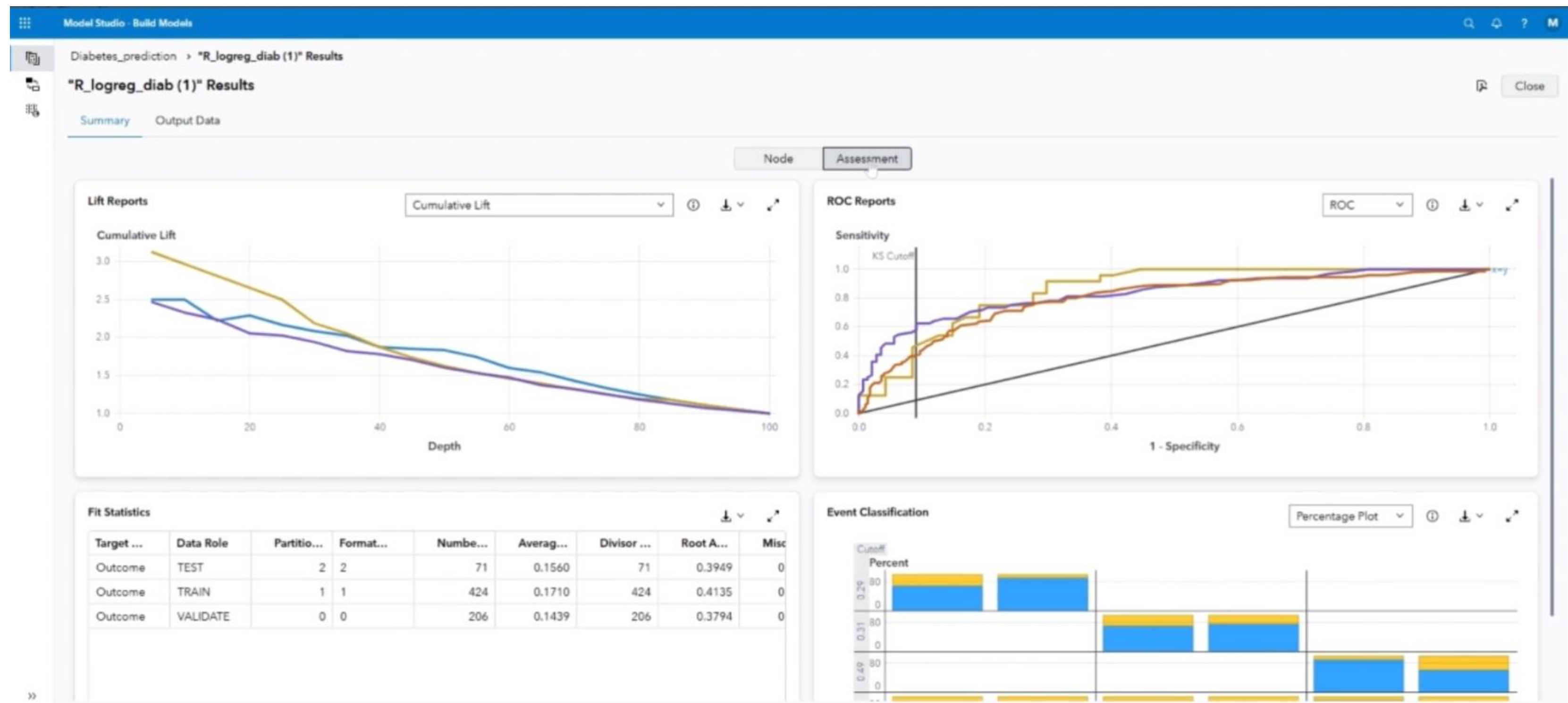
```
1 WARNING: ignoring environment value of R_HOME
2
3 Call:
4 glm(formula = dm_model_formula, family = binomial(link = "logit"),
5     data = dm_traindf)
6
7 Coefficients:
8             Estimate Std. Error z value Pr(>|z|)
9 (Intercept)    -5.6876724   0.6216097  -9.150 < 2e-16 ***
10 DiabetesPedigreeFunction  0.0006853   0.0003304   1.832  0.0669 .
11 Glucose           0.0312717   0.0044204   7.187 6.61e-13 ***
12 Insulin          -0.0015447   0.0011355  -1.360  0.1737
13 Pregnancies       0.1613106   0.0402453   4.008 6.12e-05 ***
14 SkinThickness     0.0172722   0.0084318   2.048  0.0405 *
15 ---
16 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
17
```

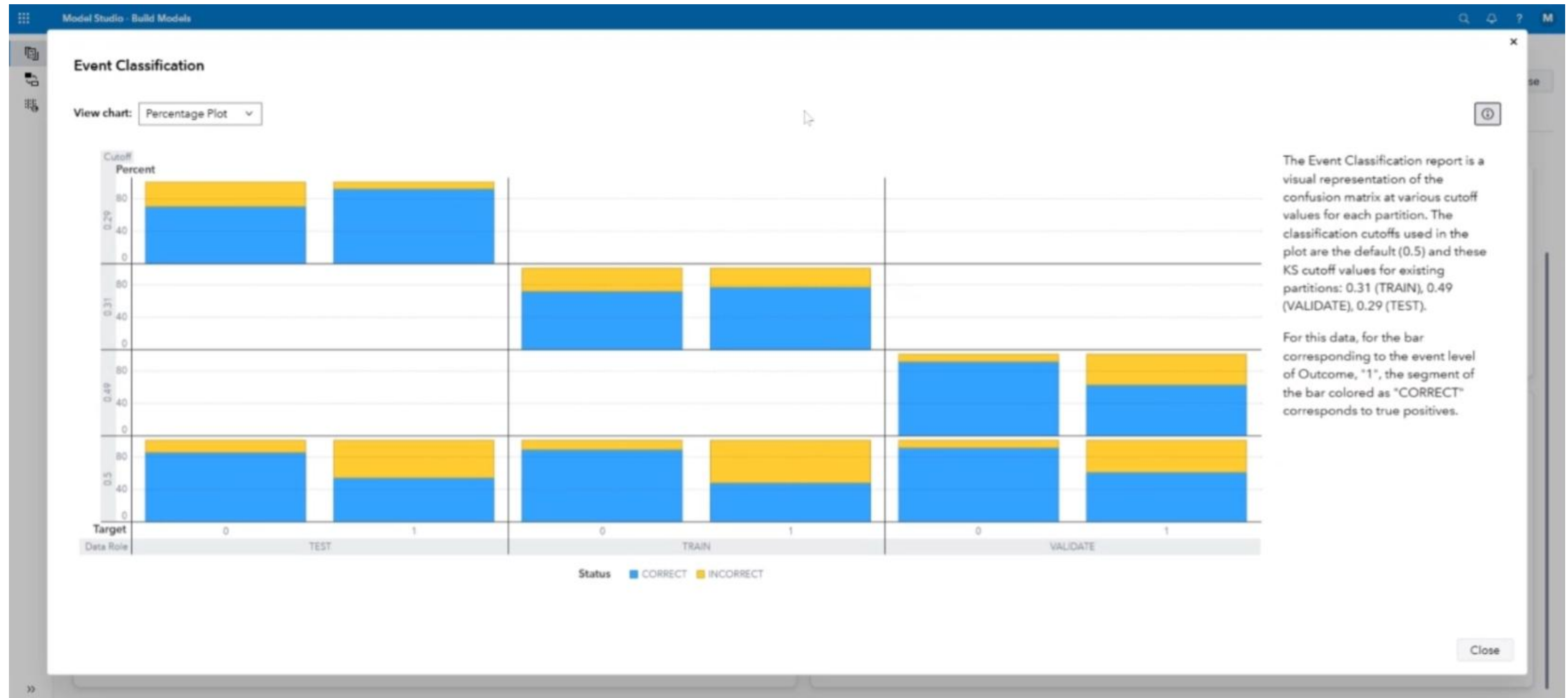
Properties

Property Name	Property Value
codeLocation	mlearning
dataMiningVersion	V2024.09
dframe	true
envVar	DM_PYPATH
exactPctLift	true
exportWithFormat	true

Output

Predicted Variables							
Obs	Predicted Variable Name	Variable Name	Level Frequency	Level Frequency Percent	Raw Numeric Value	Raw Character Value	Formatted Value
1	I_Outcome	Outcome	-	-	-		
2	P_Outcome0	Outcome	500	65.1042	0		0
3	P_Outcome1	Outcome	268	34.8958	1		1





Model Studio - Build Models

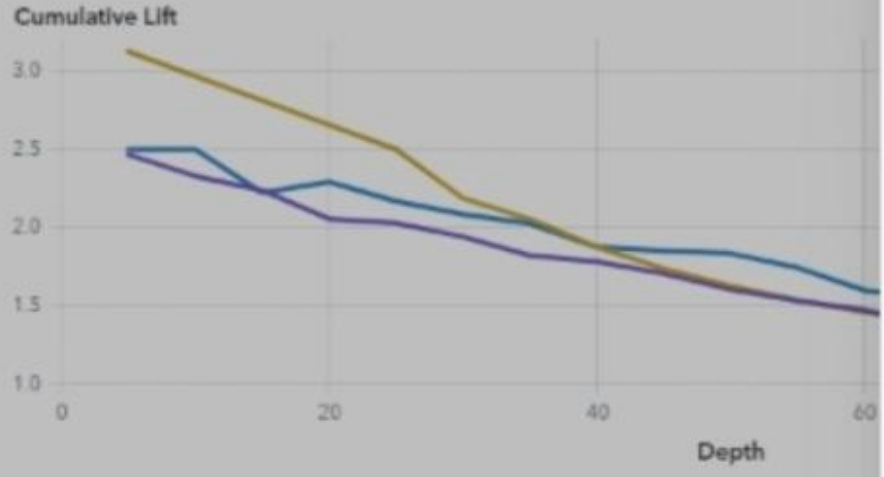
Diabetes_prediction > "R_logreg_diab (1)" Results

"R_logreg_diab (1)" Results

SummaryOutput Data

Lift Reports

Cumulative Lift



Fit Statistics

Target ...	Data Role	Partitio...	Format...	Numbe...
Outcome	TEST	2	2	71
Outcome	TRAIN	1	1	424
Outcome	VALIDATE	0	0	206

Export PDF

Page SetupSelect Objects

Page Setup

Paper size:
Letter

Orientation:
☒ Portrait
☐ Landscape

Margins

Units:
Inches

Top:
1

Right:
1

Left:
1

Bottom:
1

Options

☒ Include table of contents

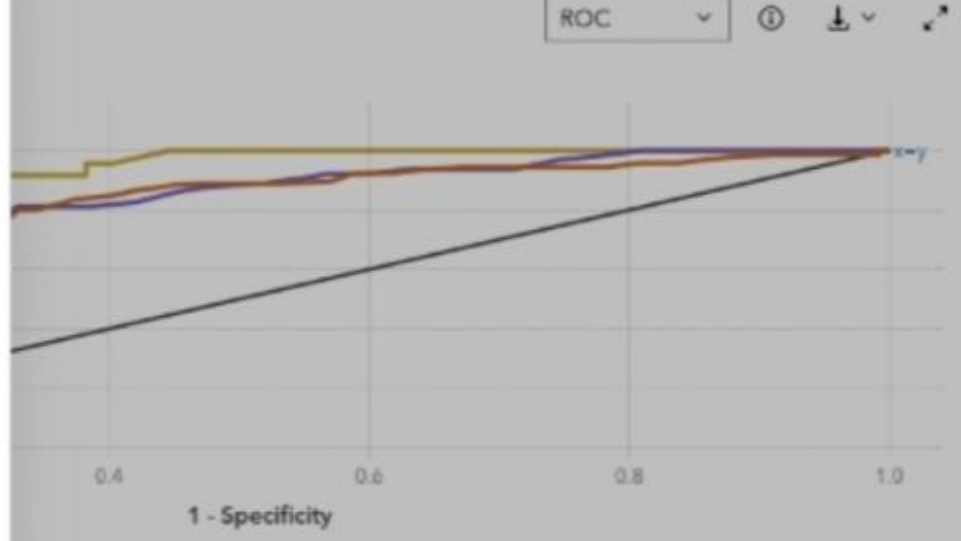
☒ Show page numbers

☒ Include title page

☒ Include details

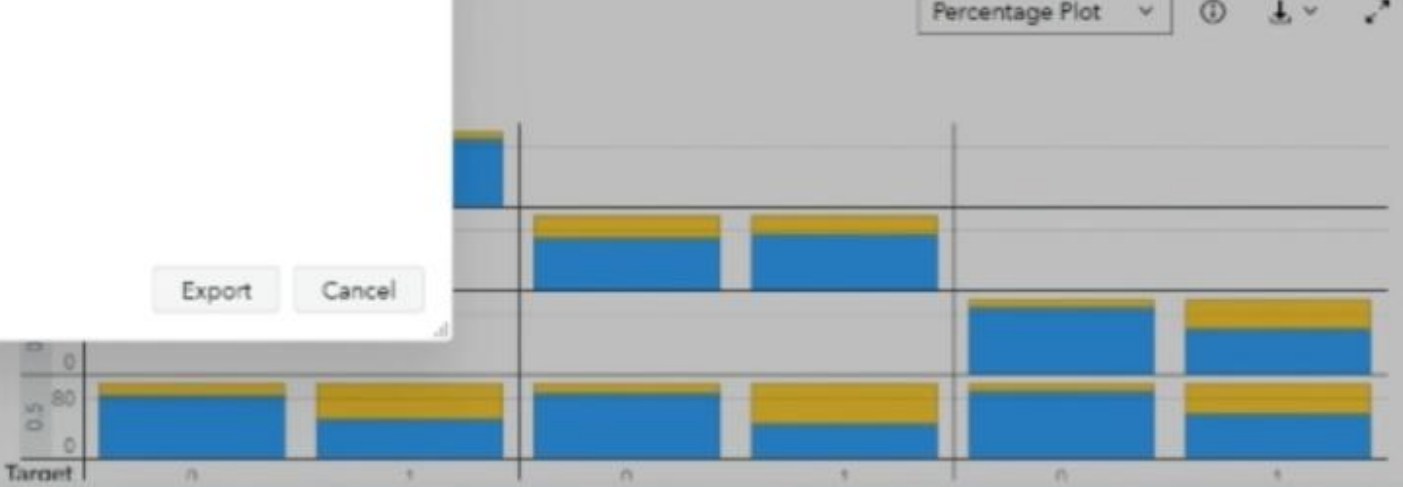
ExportCancel

ROC



1 - Specificity

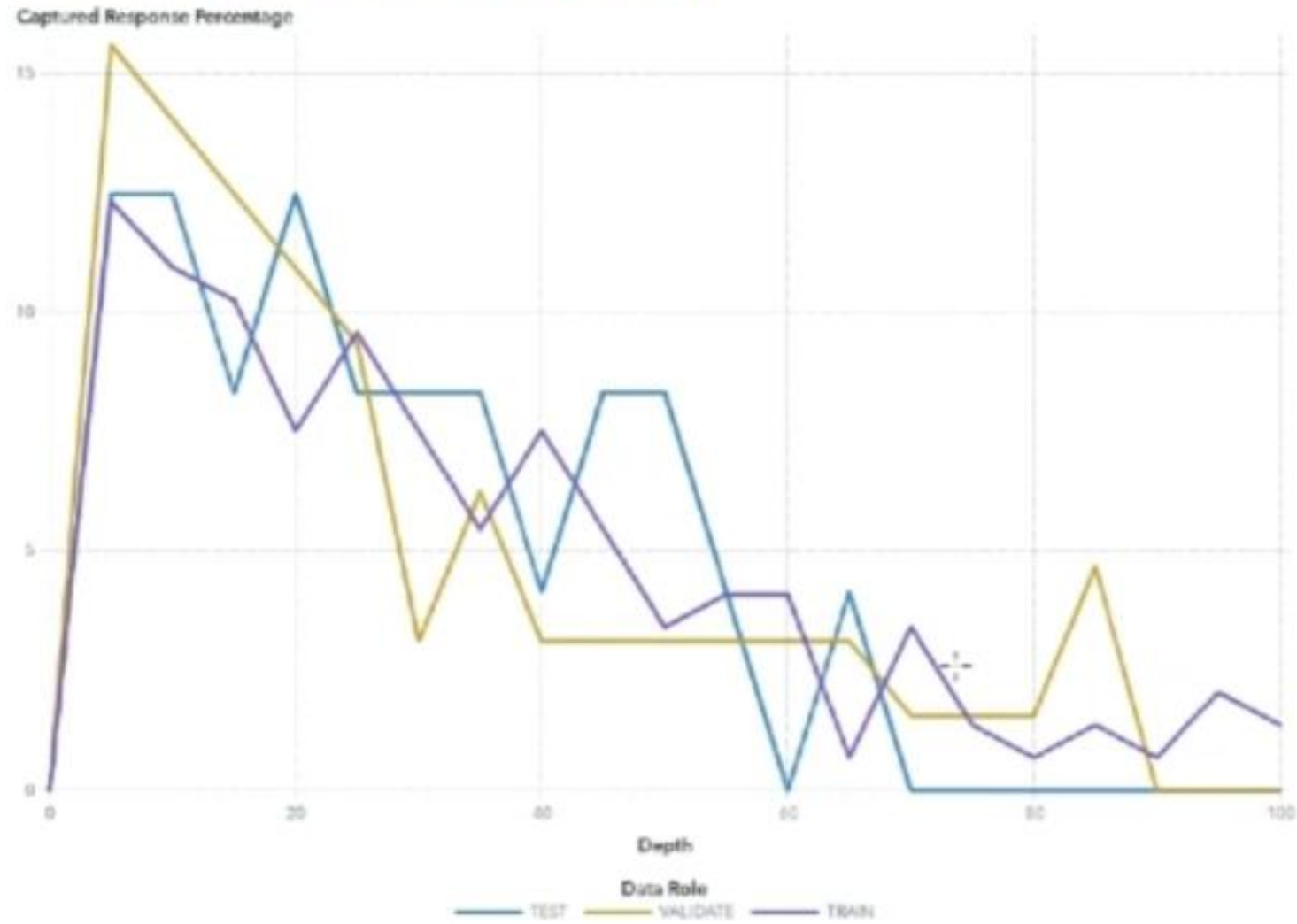
Percentage Plot



Contents

Cumulative Lift	3
Lift	5
Gain	7
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Cumulative Captured Response Percentage	11
Response Percentage	13
Cumulative Response Percentage	15
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Captured Response Percentage



At the 5% quantile (depth of 5), the VALIDATE partition has a Captured response percentage of 15.6 (compared to the expected value of 5 for no model). The best possible value of Captured response percentage for this partition at depth 5 is 17.19.

Survival Analysis



SAS Studio

vfl-032.engage.sas.com/SASStudio/

SAS Studio - Develop Code and Flows

NewOptionsViewOpenSave All

Steps

Type to filter list

SAS StepsShared

Append Table

Print Sample (final)

Survival KM Curve

Start PageFlow.flw

RunCancel

FlowGenerated CodeSubmitted Code and Results

ADTTE

Survival KM Curve

Survival KM Curve

Page1NodeNotes

Select Time Variable*

AVAL

Select Status Variable*

CNSR

Select Censored Value

0

Select Group Variable*

TRTP

Enter the units of Time

Flow.flw [temp]

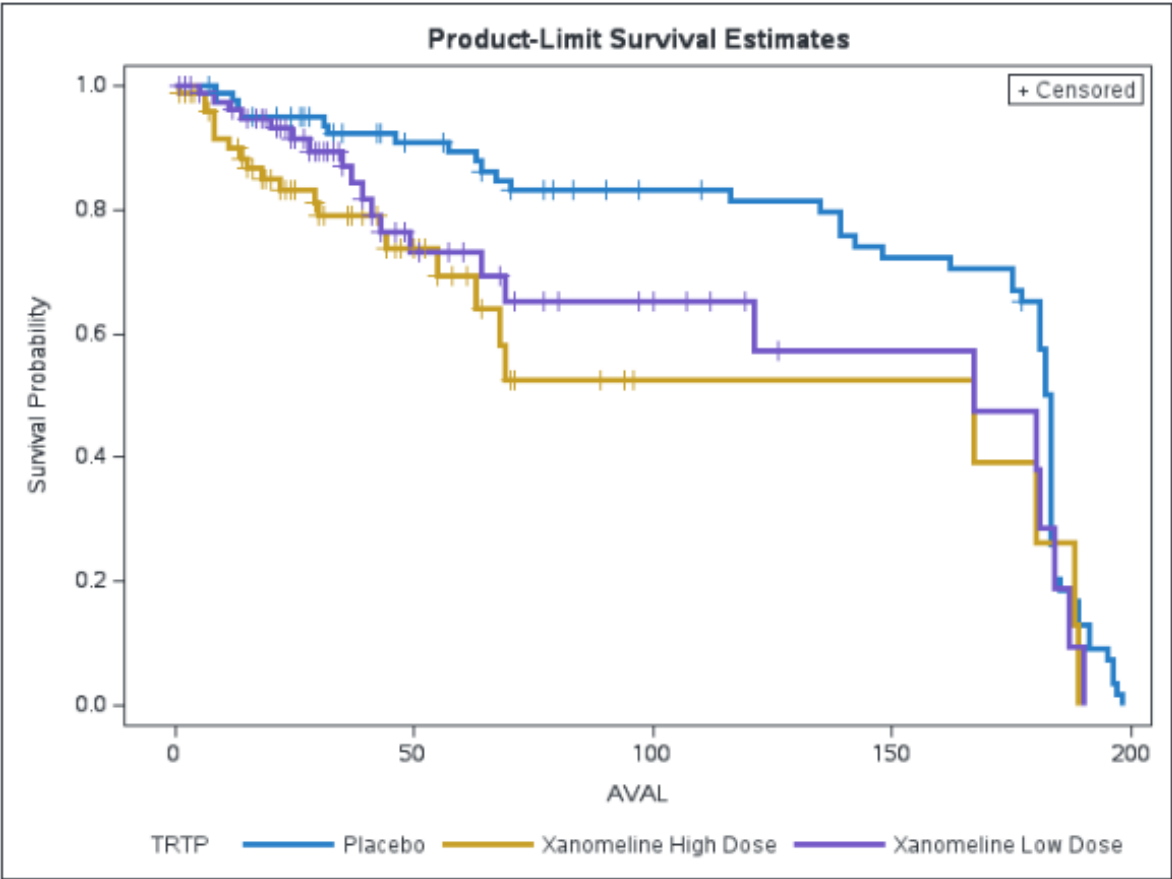
Recover (22)Submission (0)

TRTP	Log-Rank	Wilcoxon
Placebo	-12.151	-1563.0
Xanomeline High Dose	8.452	1396.0
Xanomeline Low Dose	3.699	167.0

Covariance Matrix for the Log-Rank Statistics			
TRTP	Placebo	Xanomeline High Dose	Xanomeline Low Dose
Placebo	17.0228	-7.4077	-9.6152
Xanomeline High Dose	-7.4077	10.9629	-3.5552
Xanomeline Low Dose	-9.6152	-3.5552	13.1704

Covariance Matrix for the Wilcoxon Statistics			
TRTP	Placebo	Xanomeline High Dose	Xanomeline Low Dose
Placebo	307909	-140918	-166991
Xanomeline High Dose	-140918	251039	-110120
Xanomeline Low Dose	-166991	-110120	277111

Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	9.9667	2	0.0069
Wilcoxon	10.4177	2	0.0055
-2Log(LR)	1.2968	2	0.5229



Appendix / References



For more information on

- ADTTE, please refer to the document “[Adam_tte_final_v1.pdf](#)”
- [Data Dictionary](#): Complete data description
- Custom Steps, refer to this [Video](#)
- Building [Beautiful Reports](#): Guidance on how to create a compelling data story by designing a report that is both visually appealing and easy to understand.
- SAS Solution: [SAS Clinical Acceleration](#)



Let's start the Practice

Log in to SAS

<https://learn.sas.com>

